

In this problem set, the questions marked with a $(\star)$ are required for students enrolled in 583 and bonus for students enrolled in 483.

1. Consider flat spacetime written in cylindrical coordinates

$$(t, r, \phi, z),$$

with line element

$$ds^2 = -dt^2 + dr^2 + r^2 d\phi^2 + dz^2.$$

- (a) (6 points) Coordinate Basis

1. Write down the coordinate basis vectors.
2. Compute the squared norm of each basis vector using the metric.
3. Explain why this basis is not orthonormal, even though the spacetime is flat.

- (b) (4 points) Orthonormal Basis

Define a new set of basis vectors

$$\hat{e}_t = \partial_t, \quad \hat{e}_r = \partial_r, \quad \hat{e}_\phi = \frac{1}{r} \partial_\phi, \quad \hat{e}_z = \partial_z.$$

1. Show that this basis is orthonormal.
2. Explain the physical meaning of this basis.

- (c) (6 points) Vector Components in Different Bases

Consider the vector

$$\vec{V} = V^r \partial_r + V^\phi \partial_\phi$$

with

$$V^r = 0, \quad V^\phi = \omega,$$

where ω is a constant.

1. Compute the magnitude of $\vec{V}$ using the metric.
2. Find the components of $\vec{V}$ in the orthonormal basis.
3. Explain why the orthonormal components correspond to physical measurements.

- (d) (2 points) An observer at fixed r and z moves in the ϕ direction with angular velocity ω .

1. Which basis corresponds to what the observer measures?
2. Why can coordinate components be misleading?

2. (10 points) **Euler-Lagrange** Let

$$L = \frac{1}{2} m g_{ij}(x) \dot{x}^i \dot{x}^j. \tag{1}$$

Prove that the Euler-Lagrange equations can be written as

$$\ddot{x}^i + \Gamma_{jk}^i \dot{x}^j \dot{x}^k = 0, \tag{2}$$

where $\Gamma_{jk}^i(x) = \frac{1}{2} g^{i\ell} (\partial_k g_{\ell j} + \partial_j g_{\ell k} - \partial_\ell g_{jk})$.

3. **Practice with Christoffel symbols** Consider the geometry of a 2-sphere of radius R . We define the usual coordinate system on the sphere with coordinate θ as the co-latitude (i.e. $\pi/2$ -latitude) and ϕ as the longitude.

(a) (4 points) Show, using a geometric argument, that the line element is

$$ds^2 = R^2 d\theta^2 + R^2 \sin^2 \theta d\phi^2. \quad (10)$$

(b) (12 points) Find the metric tensor components $g_{\mu\nu}$ and the Christoffel symbols $\Gamma_{\alpha\beta}^\mu$ on the sphere.

4. (★) **Geodesics on the sphere**

(a) (4 points) Write the geodesic equations on the sphere in the form of $\ddot{\theta} = f(\theta, \phi, \dot{\theta}, \dot{\phi})$ and $\ddot{\phi} = g(\theta, \phi, \dot{\theta}, \dot{\phi})$, where the dot indicates a derivative with respect to arc-length.

(b) (5 points) Show from these equations that the quantities

$$\tilde{L} = R^2 \sin^2 \theta \dot{\phi} \text{ and } \mathbf{u} \cdot \mathbf{u} = R^2 (\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) \quad (16)$$

are conserved. What is the interpretation of $\tilde{L}$?

For the rest of the problem, we will take the normalization that $\mathbf{u} \cdot \mathbf{u} = 1$.

(c) (5 points) Use the result of part (b) to write an equation for $\dot{\theta}$ in terms of θ and $\tilde{L}$. Use this to show that s can be written as

$$s = \pm R \int \frac{d\theta}{\sqrt{1 - \frac{\tilde{L}^2}{R^2 \sin^2 \theta}}}. \quad (19)$$

(d) (6 points) Evaluate this integral and show that

$$\cos \theta = \sin I \cos \frac{s - C}{R}, \quad (22)$$

where $\tilde{L} = R \cos I$ and C is some constant. *hint: first do the substitution $z = R \cos \theta$ and then $z = R \sin I \cos \psi$.*

(e) (4 points) Explain, based on your 3D Euclidean intuition, why a particle moving at constant speed along a great circle on a sphere should exhibit sinusoidal motion on the z -axis. What are the interpretations of z , I , and ψ in this picture?